GHAZI UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE

PROGRAMMING FUNDAMENTALS

LAB #03

LABORATORY ASSIGNMENT

STUDENT NAME

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ROLL NUMBER

15

PROGRAM

BS Computer Science

SECTION

A

SUBMITTED TO

Sir Asad Majeed

The original handwritten solution, code screenshot, and output follow on the next pages.

Write a single cin statement that will read input into each of these variables.

char section;
double pay;
int age;
Assume that the following variables are defined:

QUESTION

Lab - 3

Day:

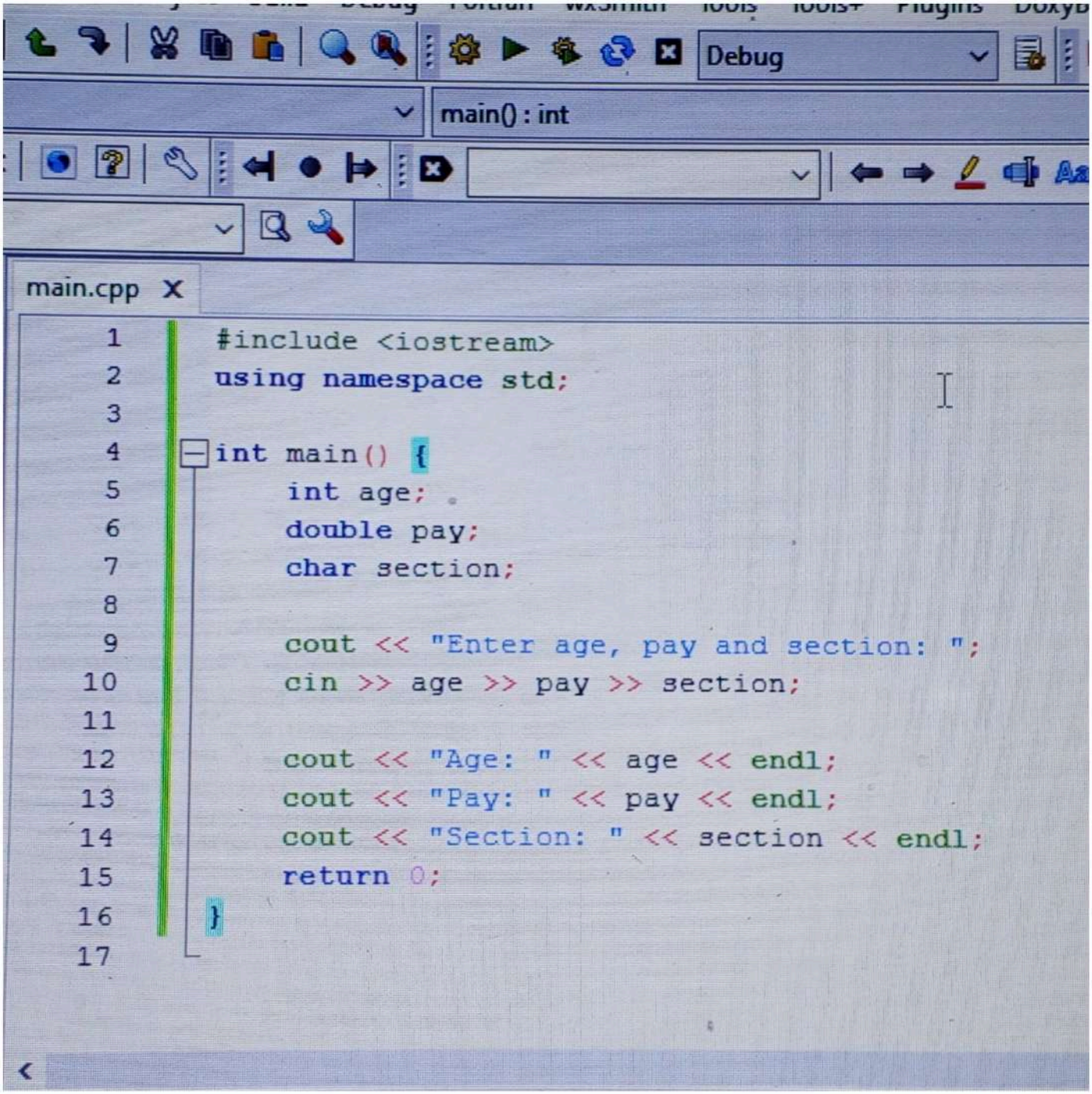
Date:

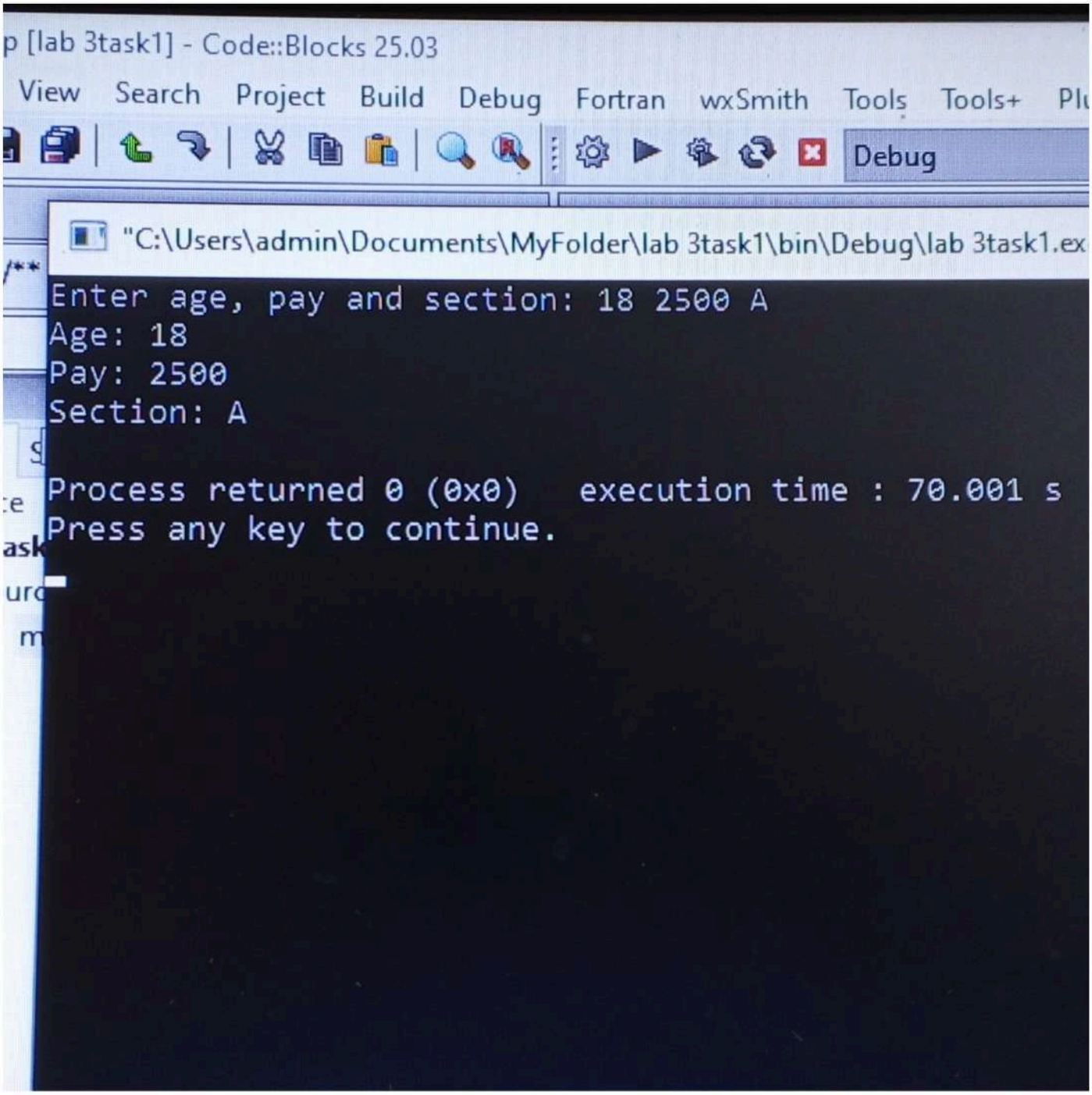
Task - 1

```
#include <iostream>
using namespace std;
int main ( ) {
    int age ;
    double pay ;
    char section;
  Cout<< "Enter age, pay and section:";
    Cin >> age >> pay >> section;
    Cout << "Age :" << age <<end l;
    Cout << "Pay:" << pay << end l;
    Cout << "section:" << section << end l;
    return 0 ;
}
```

Output :

```
Enter age, pay and section : 18 2500 A
Age : 18
Pay : 2500
section: A
```


Debug
main() : int
main.cpp
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5 int age;
6 double pay;
7 char section;
8
9 cout << "Enter age, pay and section: ";
10 cin >> age >> pay >> section;
11
12 cout << "Age: " << age << endl;
13 cout << "Pay: " << pay << endl;
14 cout << "Section: " << section << endl;
15 return 0;
16 }
17


p [lab 3task1] - Code::Blocks 25.03
View Search Project Build Debug Fortran wxSmith Tools Tools+
Debug
"C:\Users\admin\Documents\MyFolder\lab 3task1\bin\Debug\lab 3task1.ex
Enter age, pay and section: 18 2500 A
Age: 18
Pay: 2500
Section: A
Process returned 0 (0x0) execution time : 70.001 s
Press any key to continue.

The original handwritten solution, code screenshot, and output follow on the next pages.

(19 - 3) * (2 + 2) / 4
12 / (10 - 6)
(16 + 7) % 2 - 1
(8 + 7) * 2
2 + 22 * (9 - 7)
6 + 17 % 3 - 2
4 + 8 * 2
6 + 12 * 2 - 8
28 / 4 - 2

language rules:
Complete the following table by writing the value of each expression in the Value column according to C++

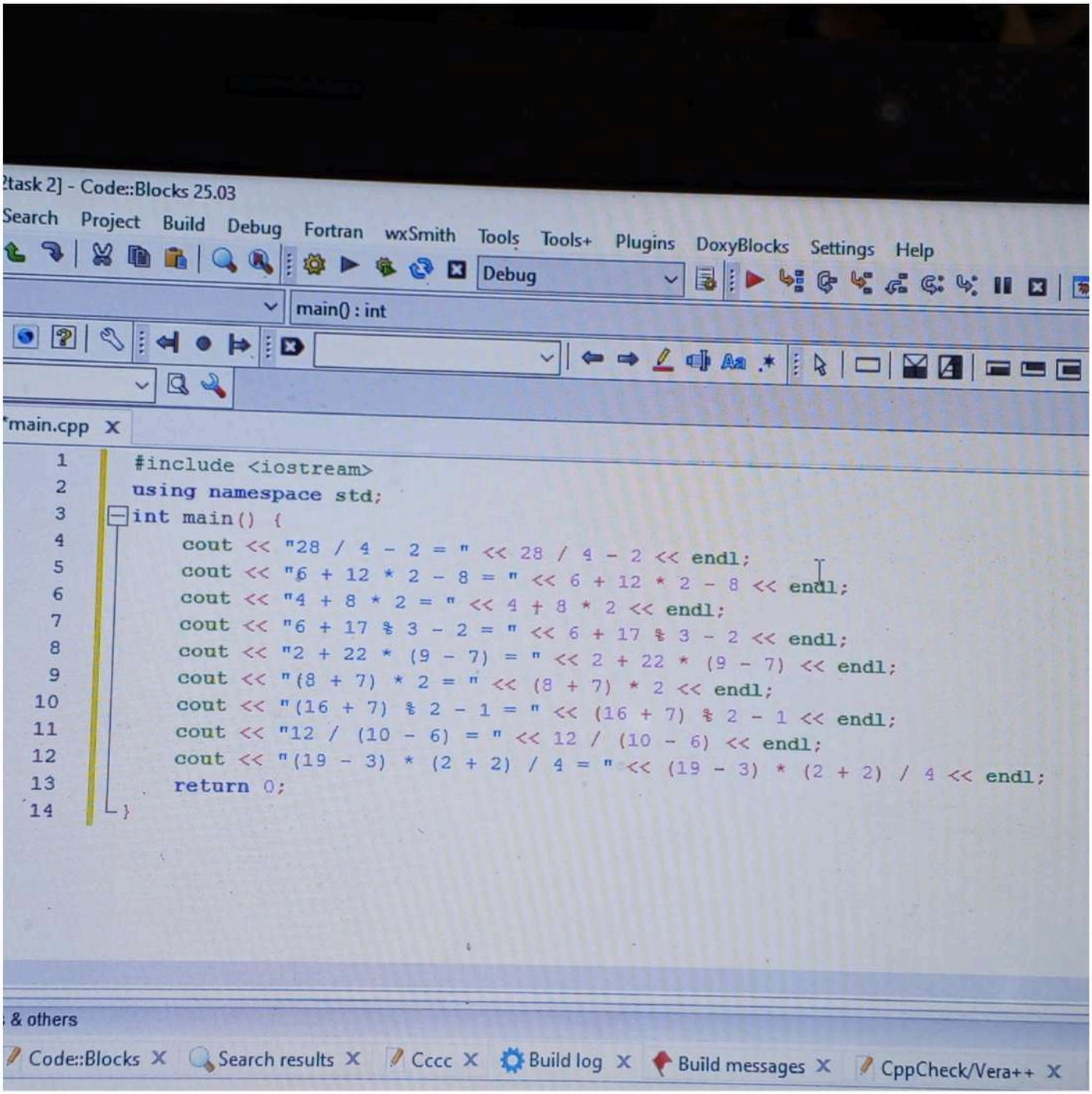
QUESTION

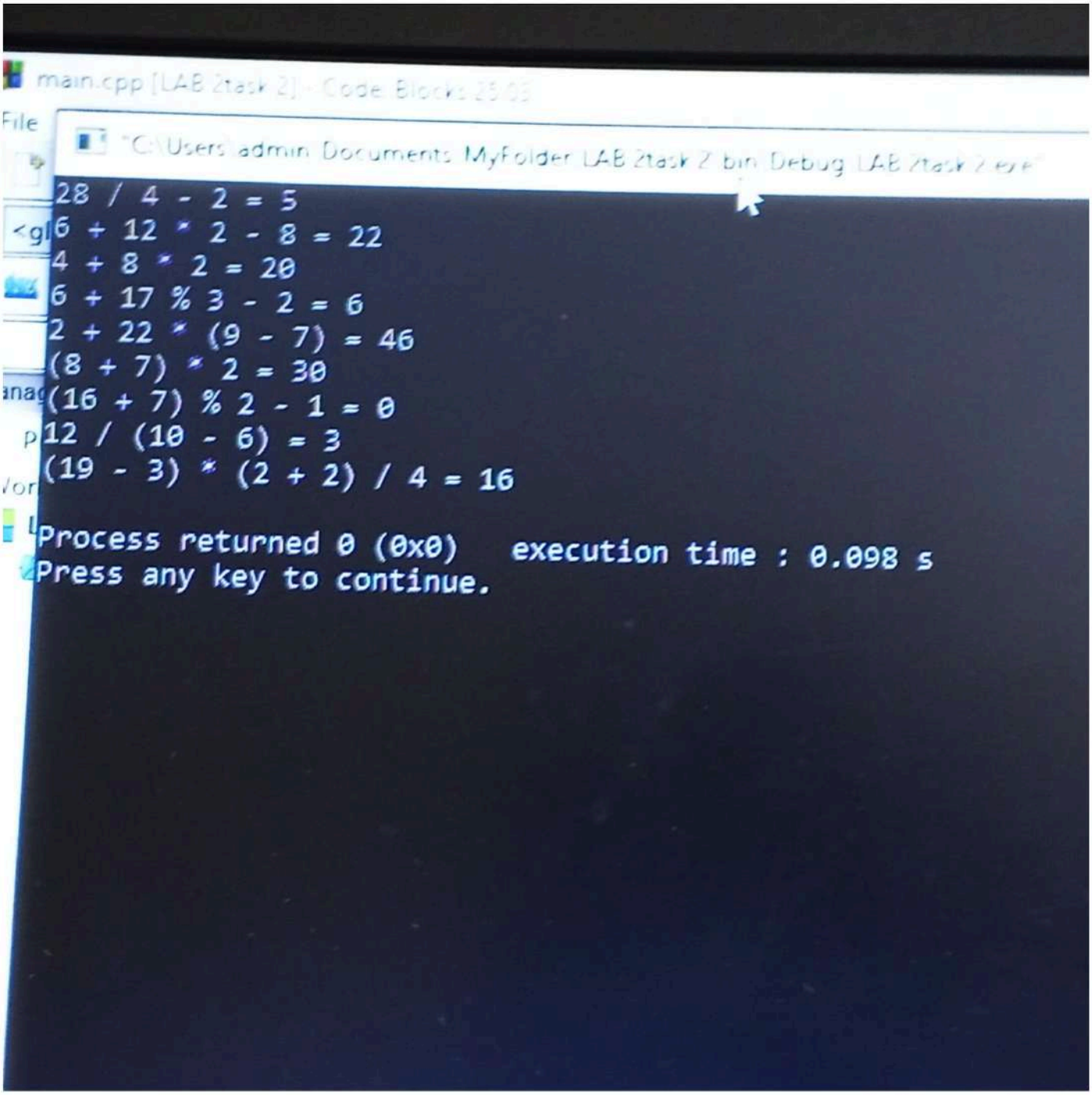
Lab - 03 Task - 02

```
#include <iostream>
Using namespace std.
int main ( )
{
    cout << " 28/4 - 2 = " << 28/4 - 2 << endl;
    cout << "6 + 12 * 2 - 8 = " << 6 + 12 * 2 - 8 << endl;
    cout << "4 + 8 * 2 = " << 4 + 8 * 2 << end l;
    cout << "6 + 17 % 3 - 2 = " << 6 + 17 % 3 - 2 << endl;
    cout << " 2 + 22 * (9 - 7) = " << 2 + 22 * (9 - 7) << endl;
    cout << "(8 + 7) * 2 = " << (8 + 7) * 2 << endl;
    cout << "(16 + 7) % 2 - 1 = " << (16 + 7) % 2 - 1 << endl;
    cout << "12 / (10 - 6) = " << 12 / (10 - 6) << endl;
    cout << "(19 - 3) * (2 + 2) / 4 = " << (19 - 3) * (2 + 2) / 4 << endl;
    return 0;
}
```

Output

```
28/4 - 2 = 5
6 + 12 * 2 - 8 = 22
4 + 8 * 2 = 20
6 + 17 % 3 - 2 = 6
2 + 22 * (9 - 7) = 46
(8 + 7) * 2 = 30
(16 + 7) % 2 - 1 = 0
12 / (10 - 6) = 3
(19 - 3) * (2 + 2) / 4 = 16
```


task 2] - Code::Blocks 25.03
Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings Help
Debug
main() : int
main.cpp
#include <iostream>
using namespace std;
int main() {
cout << "28 / 4 - 2 = " << 28 / 4 - 2 << endl;
cout << "6 + 12 * 2 - 8 = " << 6 + 12 * 2 - 8 << endl;
cout << "4 + 8 * 2 = " << 4 + 8 * 2 << endl;
cout << "6 + 17 % 3 - 2 = " << 6 + 17 % 3 - 2 << endl;
cout << "2 + 22 * (9 - 7) = " << 2 + 22 * (9 - 7) << endl;
cout << "(8 + 7) * 2 = " << (8 + 7) * 2 << endl;
cout << "(16 + 7) % 2 - 1 = " << (16 + 7) % 2 - 1 << endl;
cout << "12 / (10 - 6) = " << 12 / (10 - 6) << endl;
cout << "(19 - 3) * (2 + 2) / 4 = " << (19 - 3) * (2 + 2) / 4 << endl;
return 0;
}
& others
Code::Blocks
Search results
Cccc
Build log
Build messages
CppCheck/Vera++


main.cpp [LAB 2task 2] - Code Blocks 25.03
File
"C:\Users admin Documents MyFolder LAB 2task 2 bin Debug LAB 2task 2.exe"
28 / 4 - 2 = 5
6 + 12 * 2 - 8 = 22
4 + 8 * 2 = 20
6 + 17 % 3 - 2 = 6
2 + 22 * (9 - 7) = 46
(8 + 7) * 2 = 30
(16 + 7) % 2 - 1 = 0
12 / (10 - 6) = 3
(19 - 3) * (2 + 2) / 4 = 16
Process returned 0 (0x0) execution time : 0.098 s
Press any key to continue.

The original handwritten solution, code screenshot, and output follow on the next pages.

C. units is promoted to a float, mass remains a float, and the result is a double.
B. units is promoted to a float, mass remains a float, and the result is a float.
A. mass is demoted to an int, units remains an int, and the result is an int.
Which automatic data type conversion will take place?

```
weight = mass * units;
double weight;
float mass;
int units;
```

Assume a program has the following variable definitions:

QUESTION

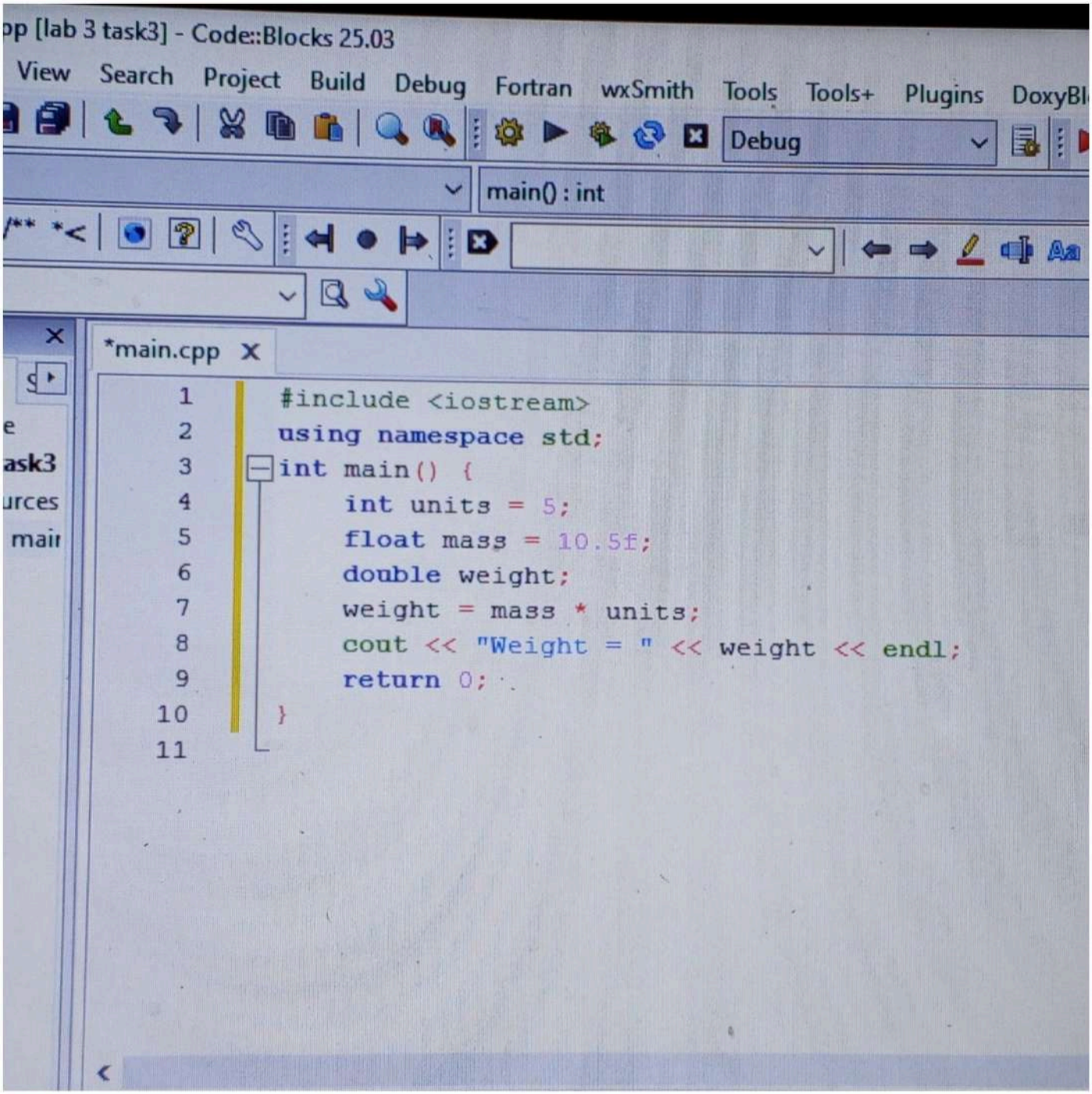
Date:

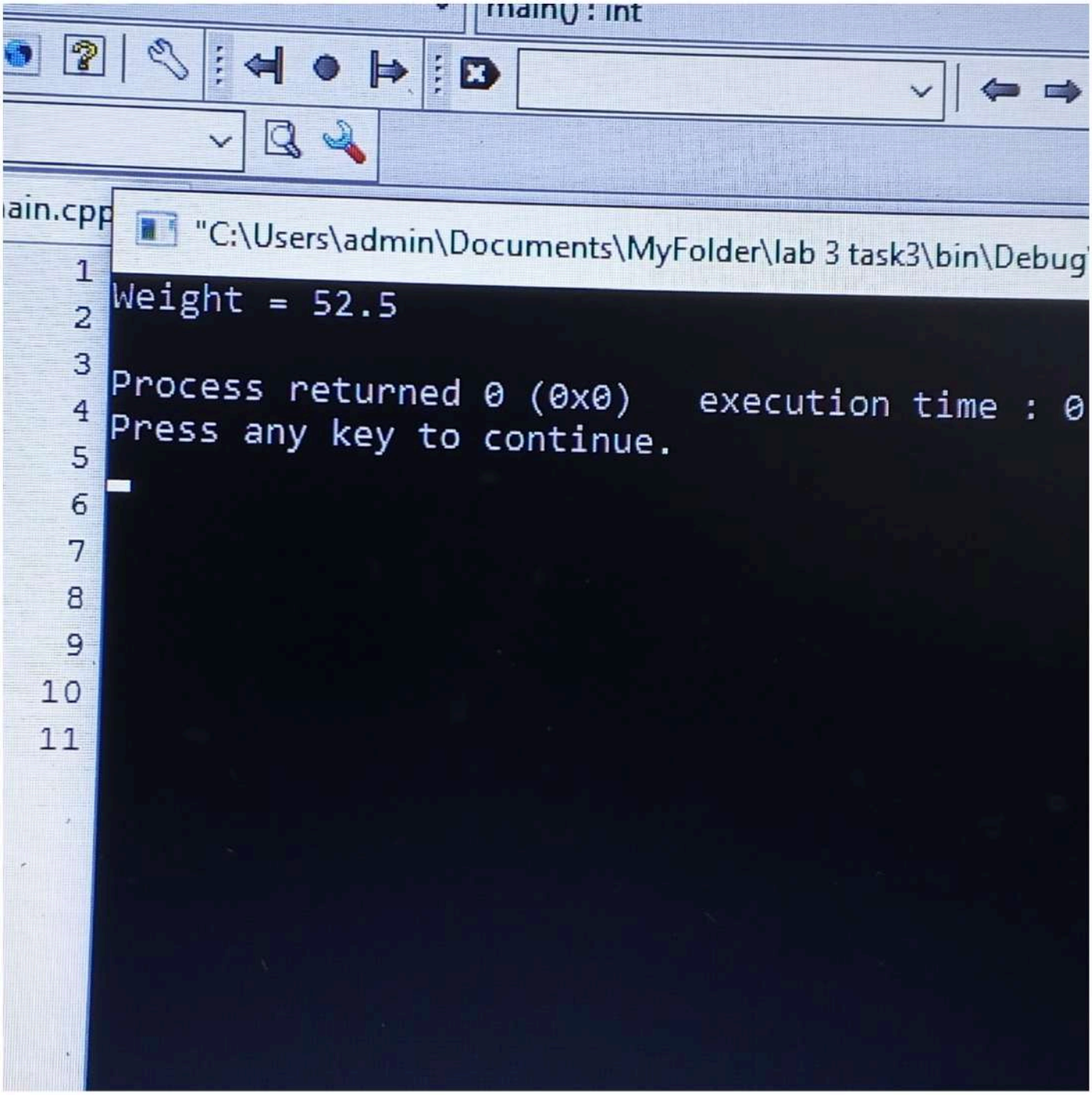
Lab-03 Task-03

```
# include <iostream>
using namespace std;
 int main() {
        int units = 5;
        Float mass = 10.5f;
        double weight ;
        weight = mass * units
       Cout << "weight = " << weight <<endl
    return 0;
}
```

output

weight = 52.5


pp [lab 3 task3] - Code::Blocks 25.03
View Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins
Debug
main() : int
*main.cpp
1 #include <iostream>
2 using namespace std;
3 int main() {
4 int units = 5;
5 float mass = 10.5f;
6 double weight;
7 weight = mass * units;
8 cout << "Weight = " << weight << endl;
9 return 0;
10 }
11


main() : int
ain.cpp
"C:\Users\admin\Documents\MyFolder\lab 3 task3\bin\Debug
1
2
3
4
5
6
7
8
9
10
11
Weight = 52.5
Process returned 0 (0x0)   execution time : 0
Press any key to continue.

The original handwritten solution, code screenshot, and output follow on the next pages.

A. 8.4 B. 8 C. 0 D. None of the above
What value will be stored in a?

and the following statement: a = b * c;

float c = 4.2;
int a, b = 2;
Assume a program has the following variable definitions:

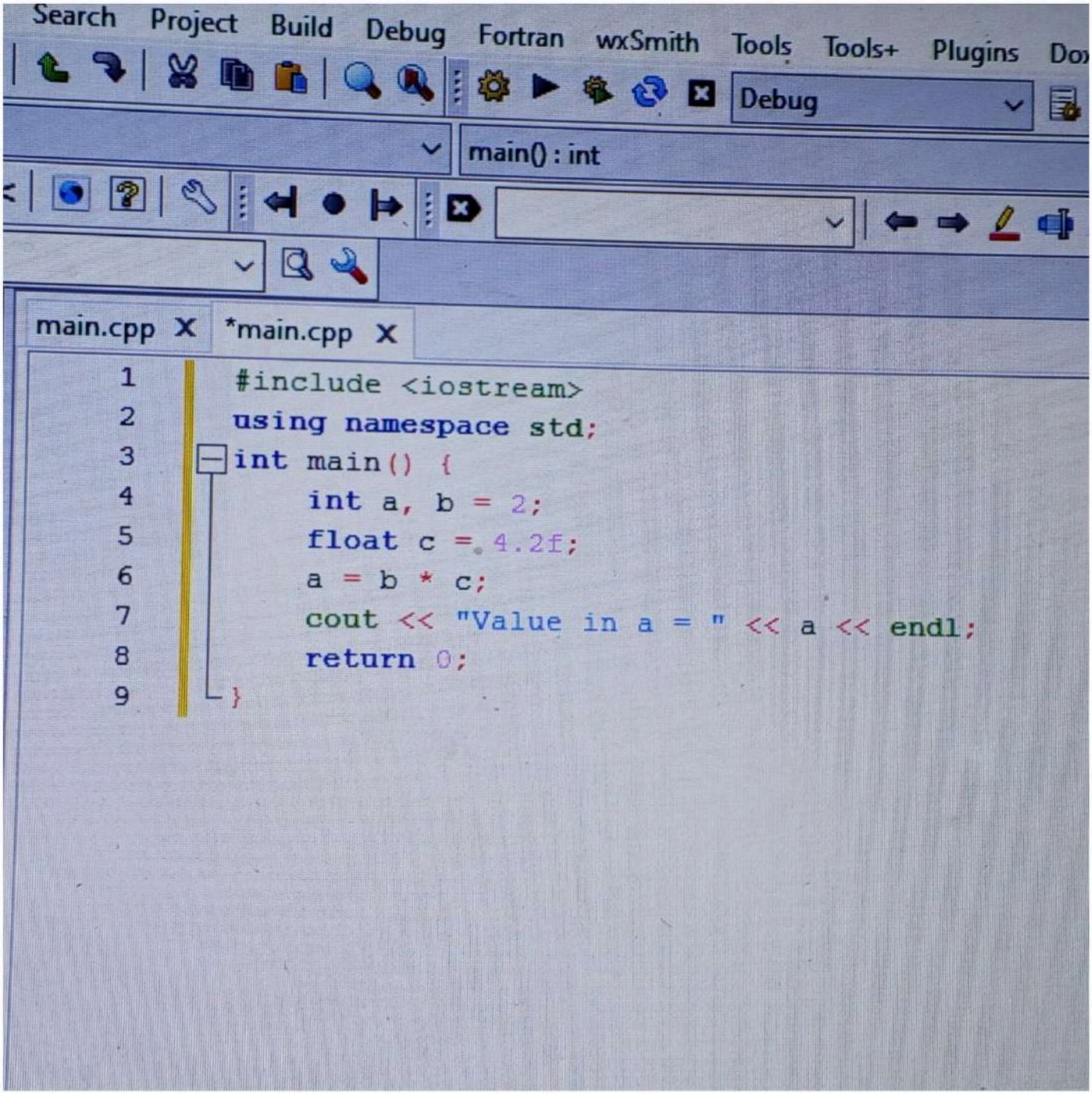
QUESTION

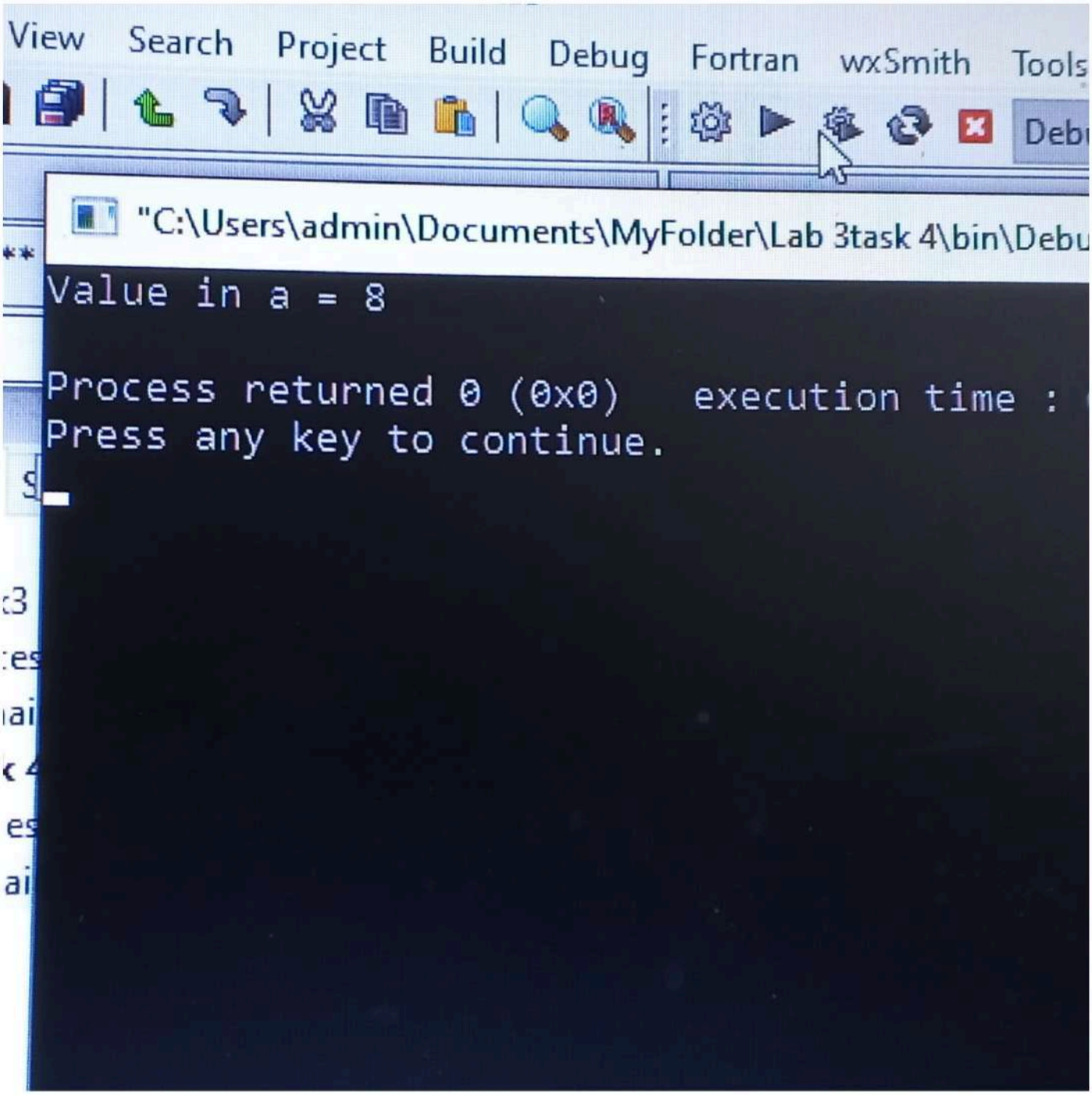
Lab-03 Task- 04

```
#include<iostream>
using namespace std;
int main() {
    int a, b = 2;
    float c = 4.2f;
    a = b * c;
    cout << "value in a = " << a << endl;
    return 0;
}
```

Output

```
Value in a = 8
```


Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins
Debug
main() : int
main.cpp
*main.cpp
#include <iostream>
using namespace std;
int main() {
int a, b = 2;
float c = 4.2f;
a = b * c;
cout << "Value in a = " << a << endl;
return 0;
}


View Search Project Build Debug Fortran wxSmith Tools
"C:\Users\admin\Documents\MyFolder\Lab 3task 4\bin\Debu
Value in a = 8
Process returned 0 (0x0) execution time :
Press any key to continue.

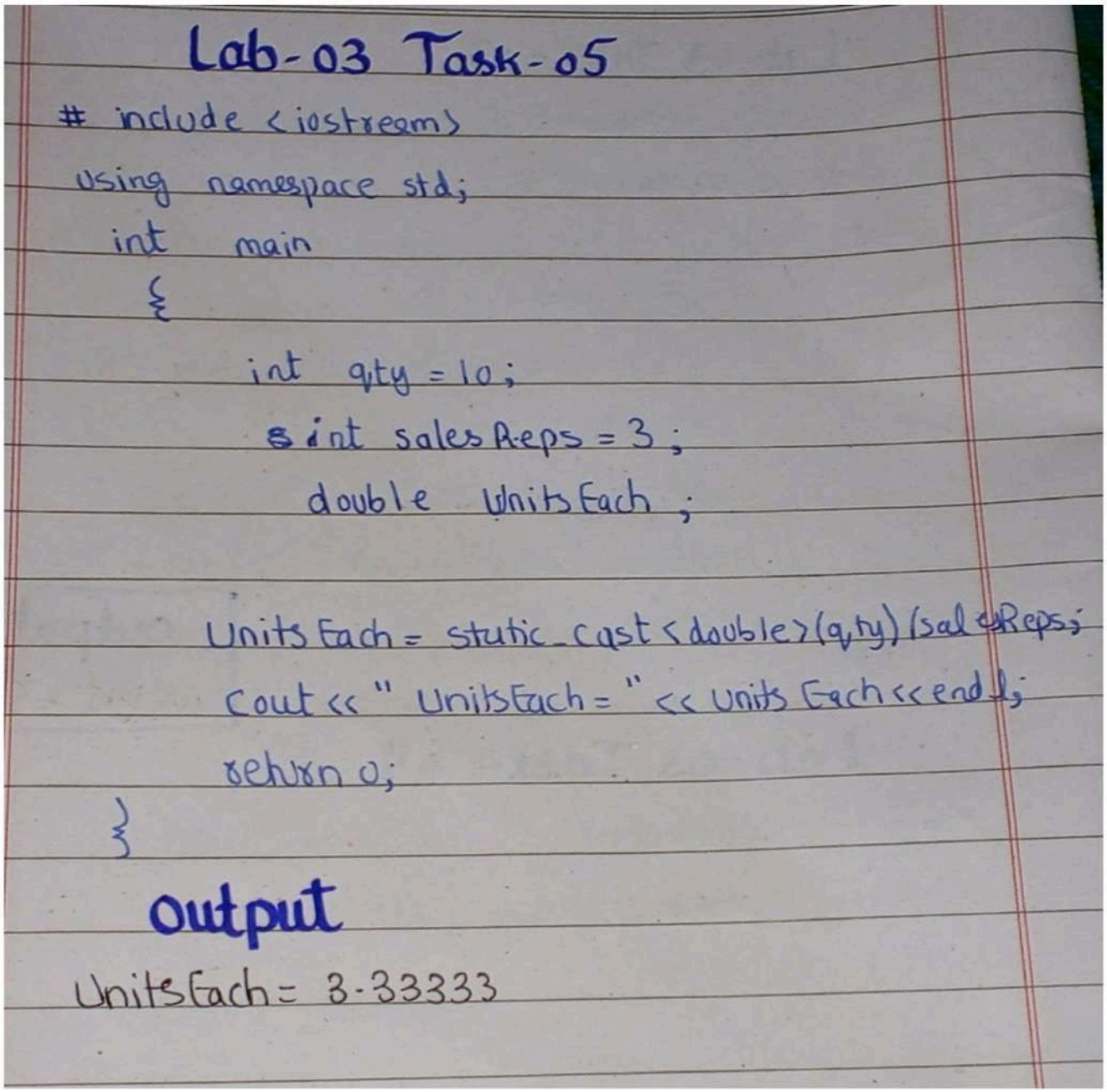
The original handwritten solution, code screenshot, and output follow on the next pages.

unitsEach = qty / salesReps;

statement so it will no longer perform integer division:
Assume that qty and salesReps are both integers. Use a type cast expression to rewrite the following

QUESTION

Lab-03 Task-05

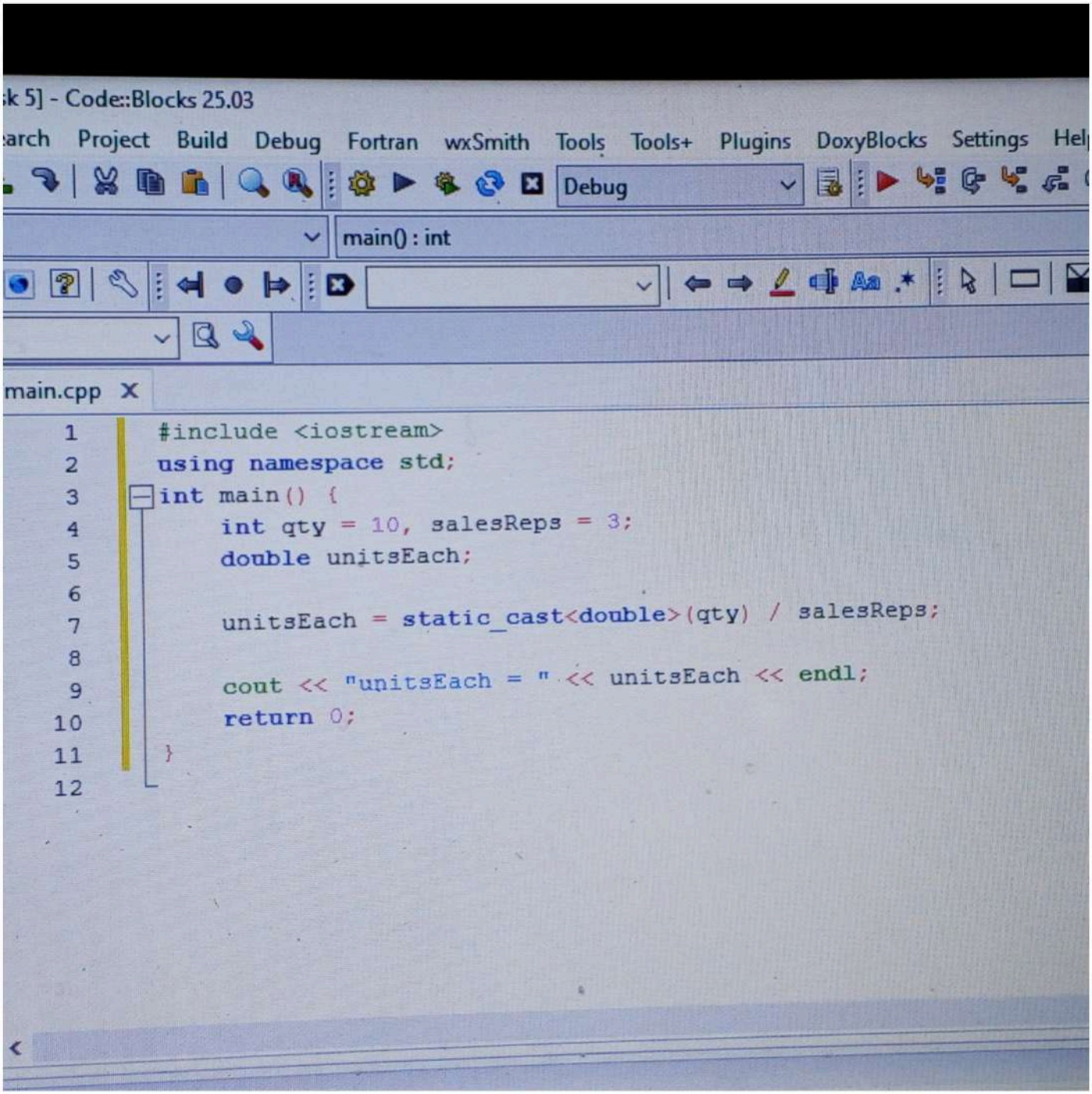


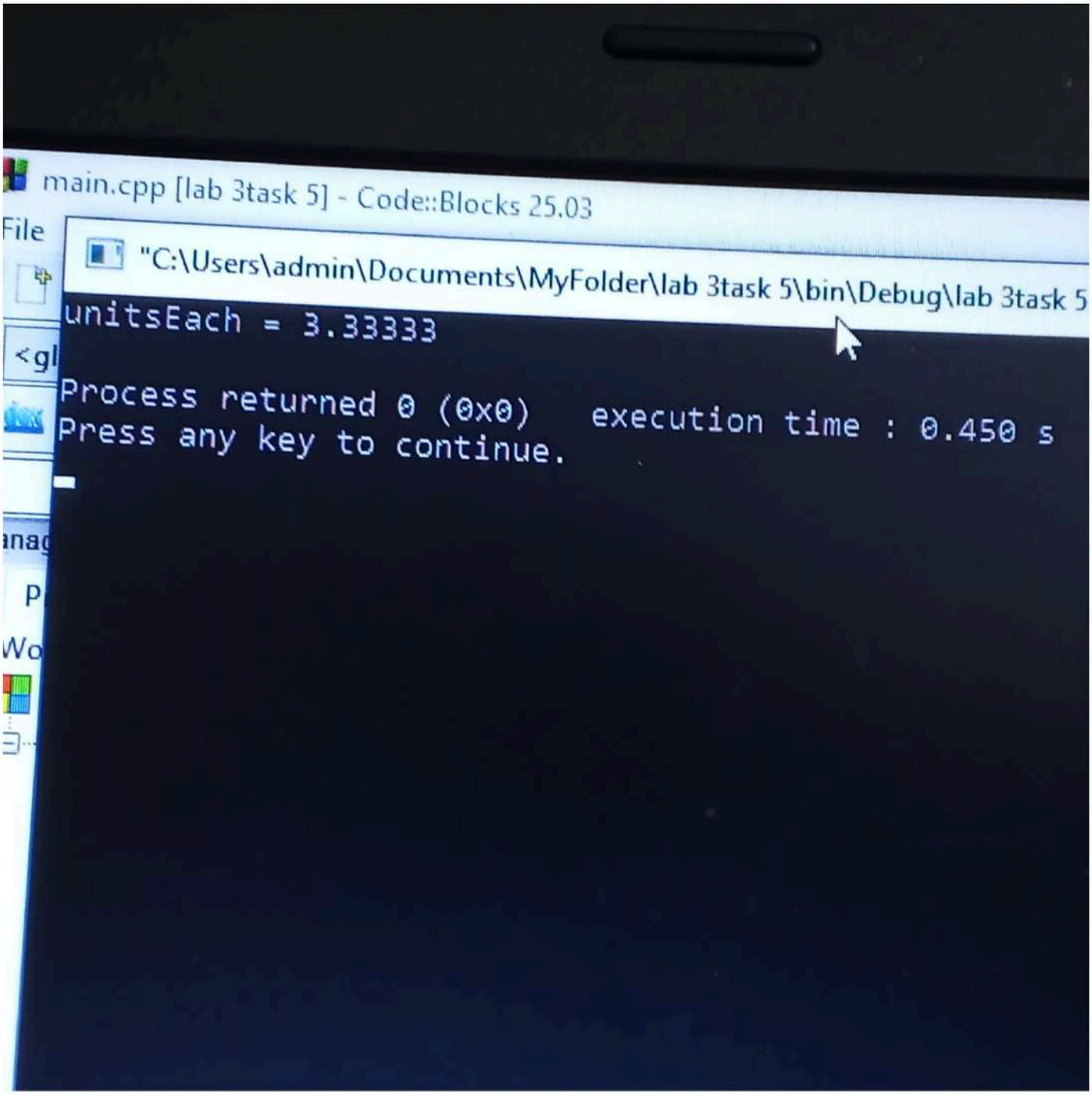
```
# include <iostream>
using namespace std;
 int  main
  {
      int qty = 10;
      int salesReps = 3;
        double UnitsEach ;

    UnitsEach = static_cast<double>(qty)/salesReps;
    cout << " UnitsEach = " << UnitsEach << endl;
    return 0;
}
```

Output

```
UnitsEach = 3.33333
```


k 5] - Code::Blocks 25.03
arch Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings Hel
Debug
main() : int
main.cpp
1 #include <iostream>
2 using namespace std;
3 int main() {
4 int qty = 10, salesReps = 3;
5 double unitsEach;
6
7 unitsEach = static_cast<double>(qty) / salesReps;
8
9 cout << "unitsEach = " << unitsEach << endl;
10 return 0;
11 }
12


main.cpp [lab 3task 5] - Code::Blocks 25.03
File
"C:\Users\admin\Documents\MyFolder\lab 3task 5\bin\Debug\lab 3task 5
unitsEach = 3.33333
Process returned 0 (0x0) execution time : 0.450 s
Press any key to continue.

The original handwritten solution, code screenshot, and output follow on the next pages.

the answer, he or she can press a key and the program will display the correct solution.
The program should then pause while the student works on the problem. When the student is ready to check

---
+ 129
247

random numbers to be added, such as:
Write a program that can be used as a math tutor for a young student. The program should display two

Math Tutor (hint rand())

QUESTION

PROGRAMMING FUNDAMENTALS · LAB #03

LAB TASK 6 · MATH TUTOR

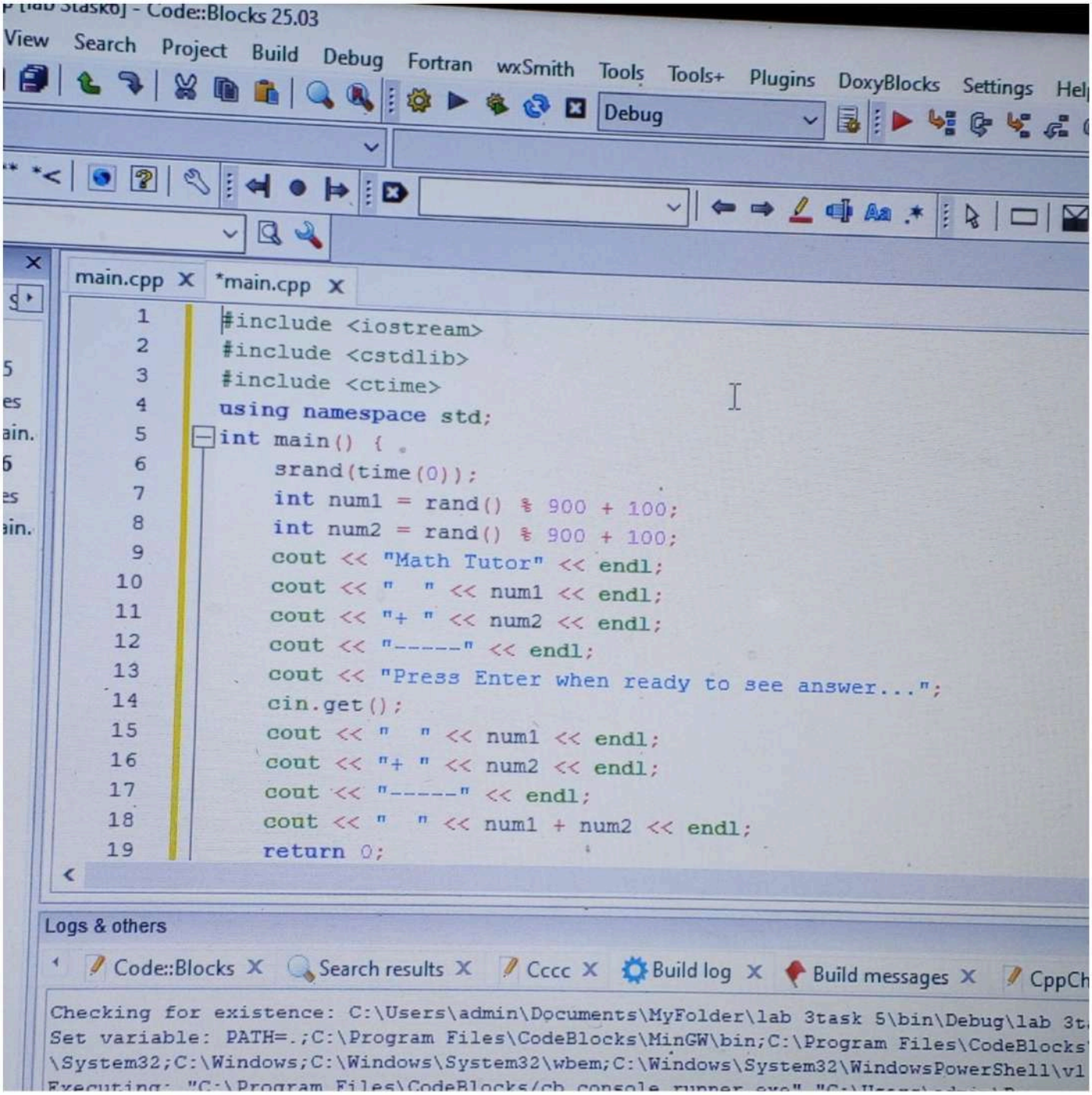
Date:

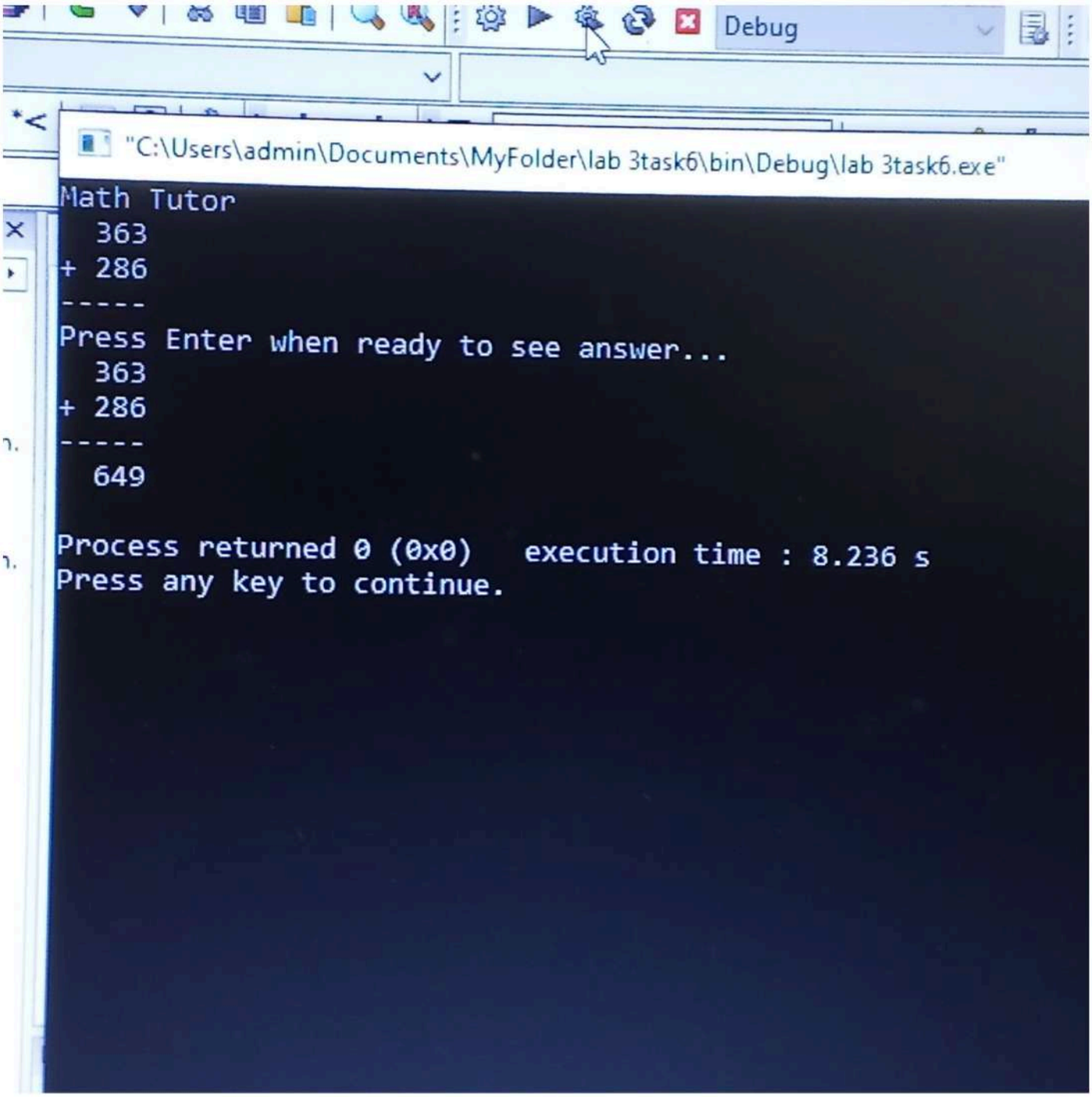
Lab-03 Task-06

```
#include <iostream>
#include <cstdlib>
#include <ctime>
Using namespace std;
int main ()
{
    srand(time(0));
    int num1 = rand()%900+100;
    int num2 = rand()%900+100;
  cout<<"Math Tutor"<<endl;
  cout<<"  "<<num1<<endl;
  cout<<"+ "<<num2<<endl;
  cout<<"-----"<<endl;
cout<<"Press Enter when ready to see answer...";
   cin.get();
  cout<<"  "<<num1<<endl;
  cout<<"+ "<<num2<<endl;
  cout<<"-----"<<endl;
  cout<<"  "<<num1+num2<<endl;
  return 0;
}
```

Output:

```
Math Tutor
  363
+ 286
.....
  363
+ 286
-----
  649
```


Code::Blocks 25.03
View Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings
Debug
main.cpp
*main.cpp
#include <iostream>
#include <cstdlib>
#include <ctime>
using namespace std;
int main() {
srand(time(0));
int num1 = rand() % 900 + 100;
int num2 = rand() % 900 + 100;
cout << "Math Tutor" << endl;
cout << " " << num1 << endl;
cout << "+ " << num2 << endl;
cout << "-----" << endl;
cout << "Press Enter when ready to see answer...";
cin.get();
cout << " " << num1 << endl;
cout << "+ " << num2 << endl;
cout << "-----" << endl;
cout << " " << num1 + num2 << endl;
return 0;
Logs & others
Code::Blocks
Search results
Cccc
Build log
Build messages
Checking for existence: C:\Users\admin\Documents\MyFolder\lab 3task 5\bin\Debug\lab 3t
Set variable: PATH=.;C:\Program Files\CodeBlocks\MinGW\bin;C:\Program Files\CodeBlocks
\System32;C:\Windows;C:\Windows\System32\wbem;C:\Windows\System32\WindowsPowerShell\v1


Debug
"C:\Users\admin\Documents\MyFolder\lab 3task6\bin\Debug\lab 3task6.exe"
Math Tutor
363
+ 286

Press Enter when ready to see answer...
363
+ 286

649
Process returned 0 (0x0) execution time : 8.236 s
Press any key to continue.

The original handwritten solution, code screenshot, and output follow on the next pages.

Program 1 is intended to input two numbers, calculate their sum, and display the sum.

Each of the following programs has some errors. Locate as many as you can.

QUESTION

PROGRAMMING FUNDAMENTALS · LAB #03

HOME TASK 1 · PROGRAM 1

ay: Date:

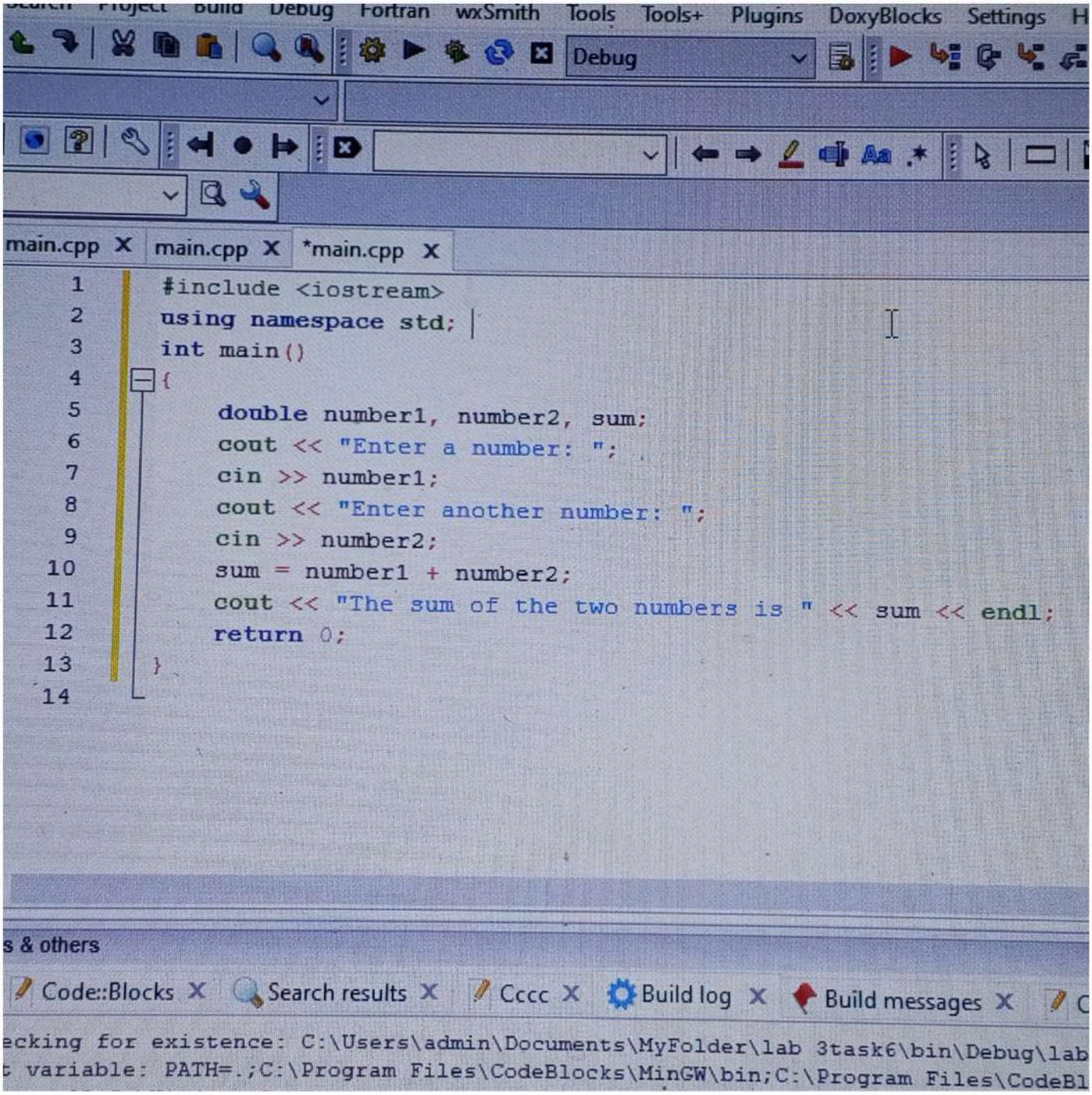
Lab-03 Home Task 1

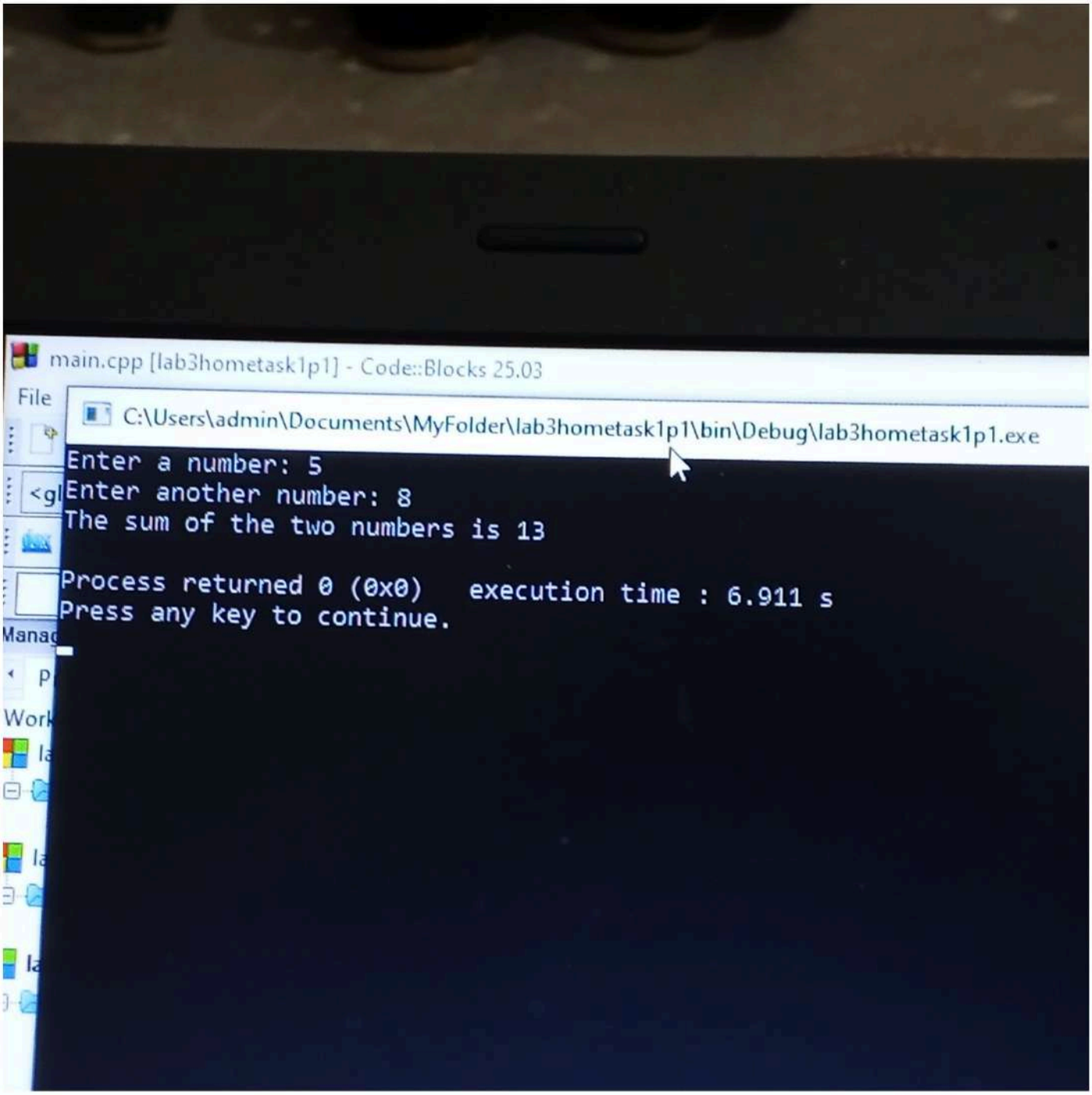
Program 1:

```
#include<iostream>
using namespace std;
int main() {
    double number1, number2, sum;
    cout<<"Enter a number: ";
    cin >> number1;
    cout<<" Enter another number: ";
    cin >> number2;
    sum = number1 + number2;
cout<<"The sum of the two numbers is "<<sum<<endl;
    return 0;
}
```

Output

```
Enter a number: 5
Enter another number: 8
The sum of the two number is 13
```


Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings
Debug
main.cpp
main.cpp
*main.cpp
#include <iostream>
using namespace std;
int main()
{
double number1, number2, sum;
cout << "Enter a number: ";
cin >> number1;
cout << "Enter another number: ";
cin >> number2;
sum = number1 + number2;
cout << "The sum of the two numbers is " << sum << endl;
return 0;
}
s & others
Code::Blocks
Search results
Cccc
Build log
Build messages
ecking for existence: C:\Users\admin\Documents\MyFolder\lab 3task6\bin\Debug\lab
t variable: PATH=.;C:\Program Files\CodeBlocks\MinGW\bin;C:\Program Files\CodeBl


main.cpp [lab3hometask1p1] - Code::Blocks 25.03
File
C:\Users\admin\Documents\MyFolder\lab3hometask1p1\bin\Debug\lab3hometask1p1.exe
Enter a number: 5
Enter another number: 8
The sum of the two numbers is 13
Process returned 0 (0x0) execution time : 6.911 s
Press any key to continue.

The original handwritten solution, code screenshot, and output follow on the next pages.

Program 2 is intended to input two integers and display their quotient using a floating-point calculation.

Each of the following programs has some errors. Locate as many as you can.

QUESTION

PROGRAMMING FUNDAMENTALS · LAB #03

HOME TASK 1 · PROGRAM 2

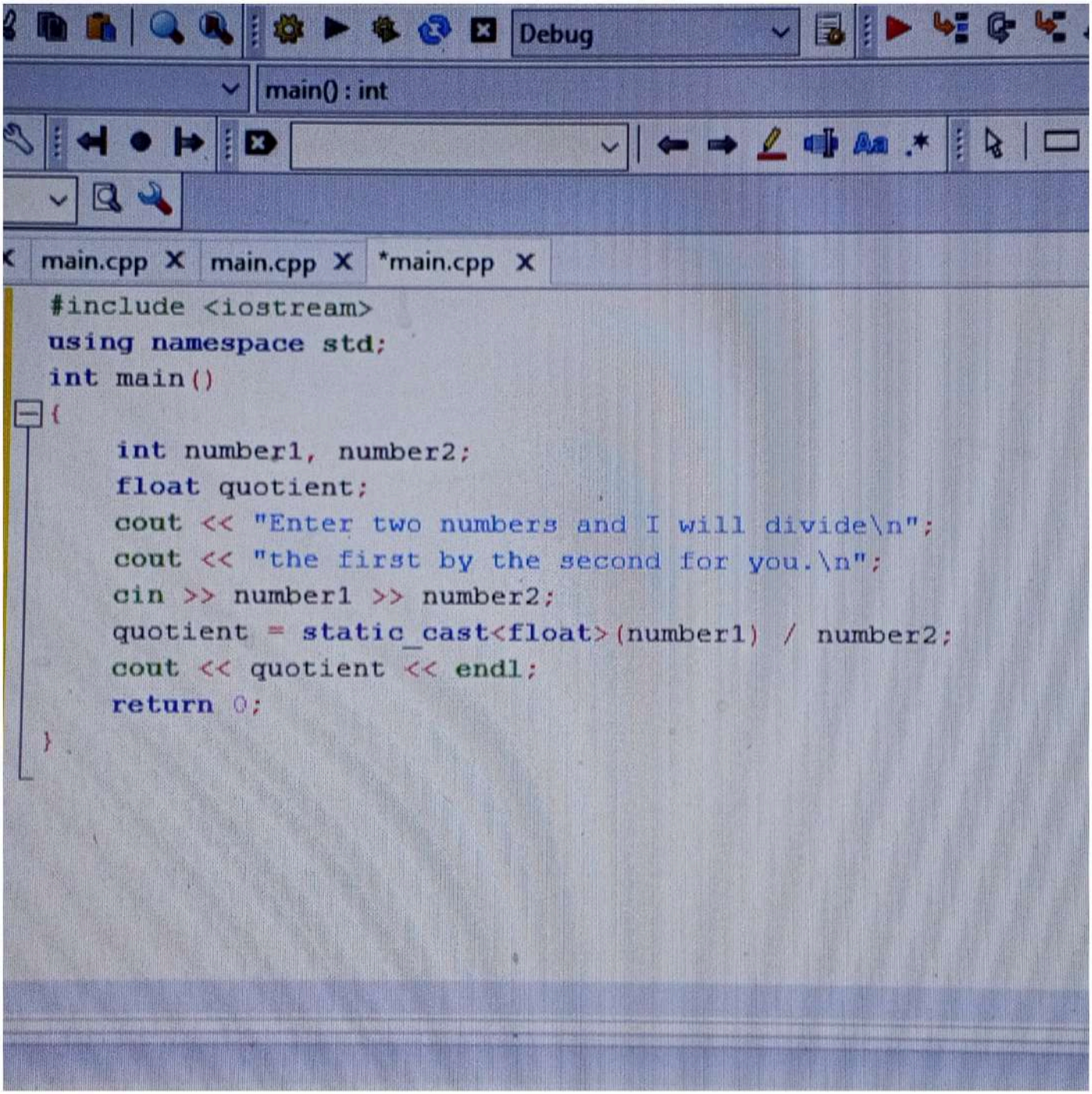
Home Task- 01

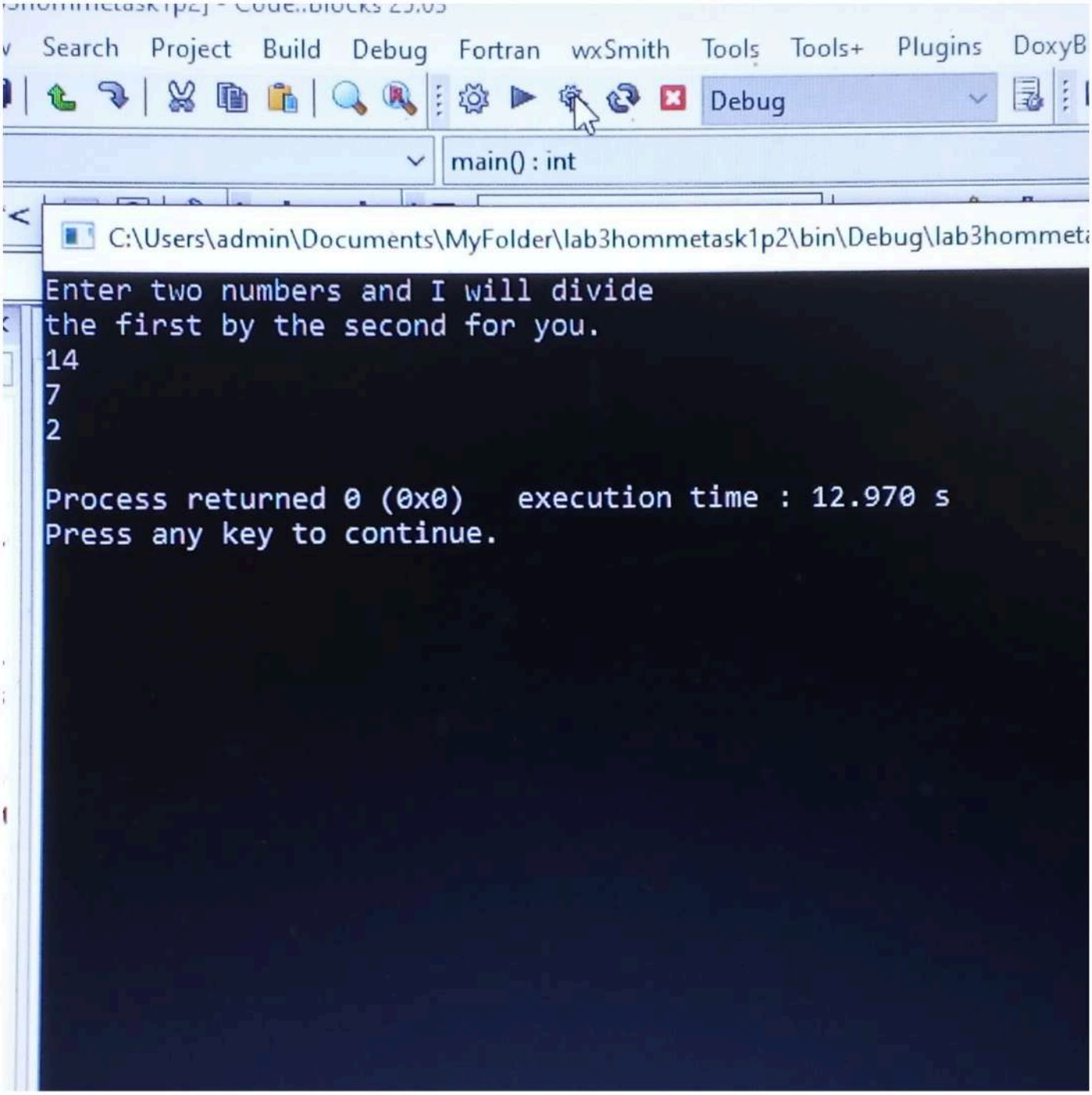
Program 2:-

```
#include <iostream>
Using namespace std;
int main()
{
    int number 1, number 2;
     float quotient ;
    cout << " Enter two numbers and I will divide\n";
   cout << " the first by the second for you.\n";
    cin >> number 1 >> number 2;
     quotient = static_cast<float>(number1)/ number 2;
     cout << quotient << endl;
     return 0;
}
```

Output

```
Enter two numbers and I will divide
the first by the second for you
14
7
2
```


Debug
main() : int
main.cpp
main.cpp
*main.cpp
#include <iostream>
using namespace std;
int main()
{
int number1, number2;
float quotient;
cout << "Enter two numbers and I will divide\n";
cout << "the first by the second for you.\n";
cin >> number1 >> number2;
quotient = static_cast<float>(number1) / number2;
cout << quotient << endl;
return 0;
}


Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyB
Debug
main() : int
C:\Users\admin\Documents\MyFolder\lab3hommetask1p2\bin\Debug\lab3hommeta
Enter two numbers and I will divide
the first by the second for you.
14
7
2
Process returned 0 (0x0) execution time : 12.970 s
Press any key to continue.

The original handwritten solution, code screenshot, and output follow on the next pages.

QUESTION

Write a program that stores the following values in five different variables: 28, 32, 37, 24, and 33. To get the average of a series of values, add the values up and then divide the sum by the number of values. The program should first calculate the sum of these five variables and store the result in a separate variable named sum. Then, divide the sum variable by 5 to get the average. Display the average on the screen.

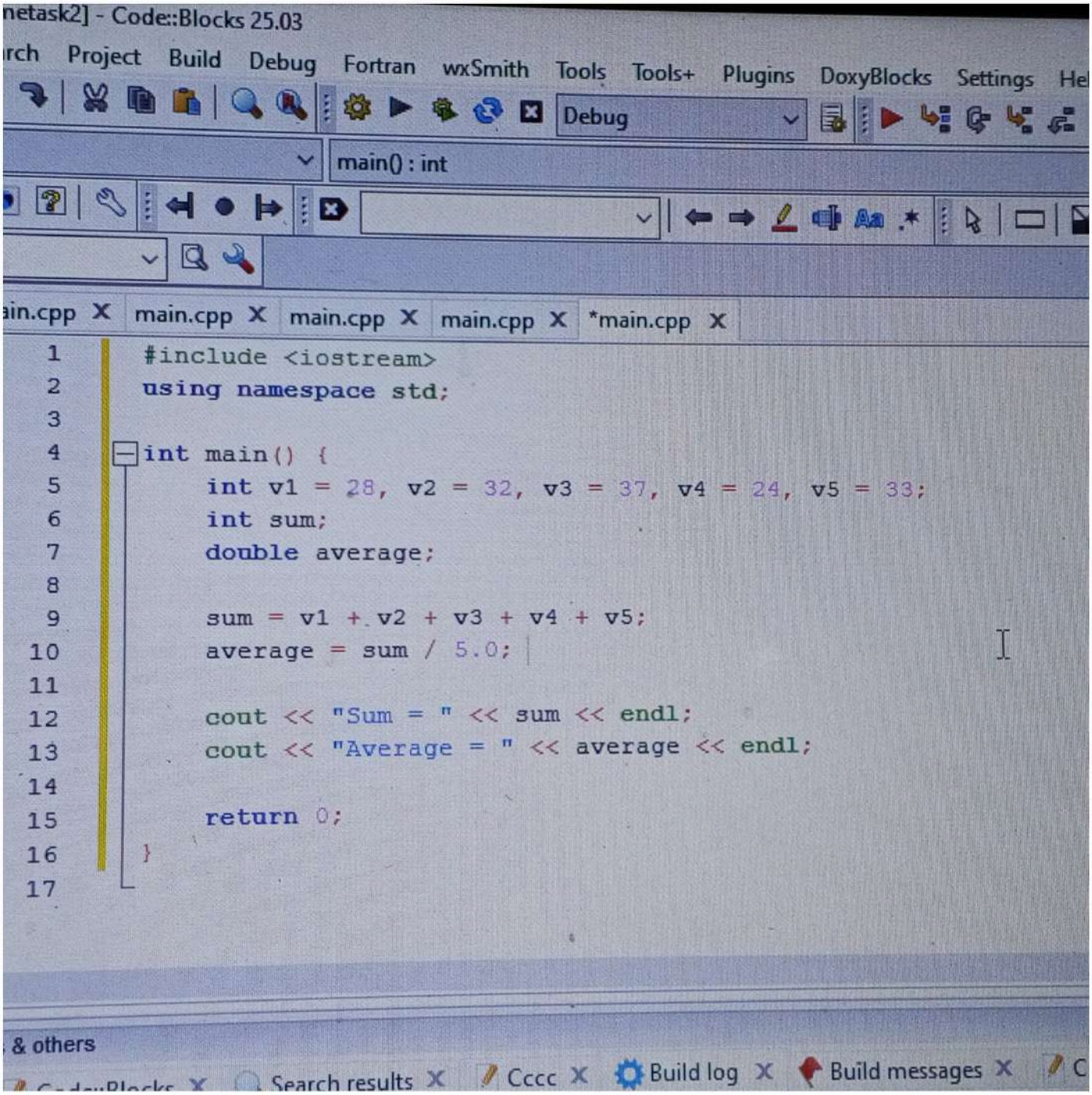
Lab-03 Home task 2

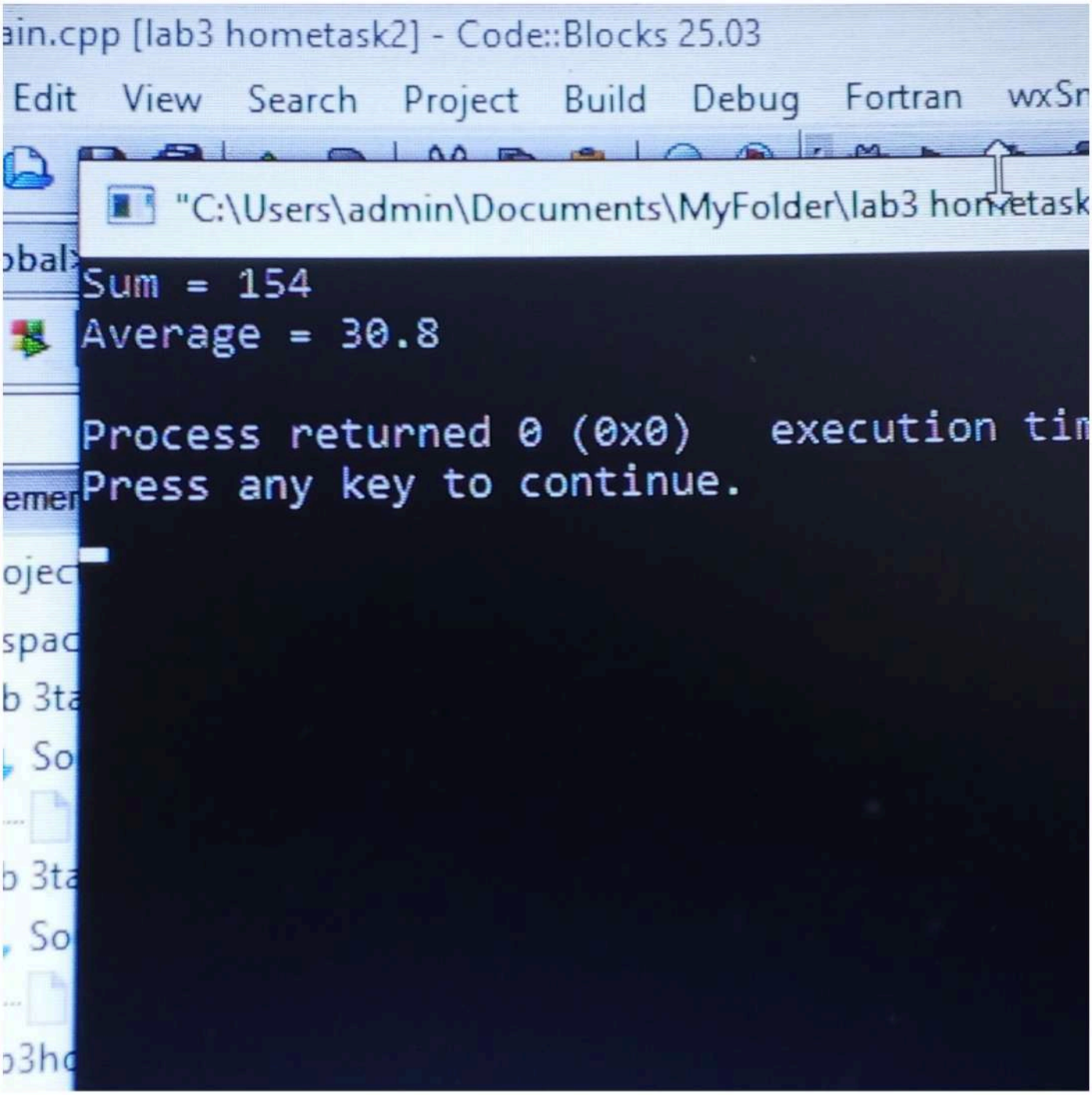
Date:

```
#include<iostream>
using namespace std;
int main () {
    int v1=28, v2=32, v3=37, v4=24, v5=33;
    int sum ;
    double average;
    sum = v1 + v2 + v3 + v4 + v5;
    average = sum / 5.0;
    cout << "sum= " << sum <<endl;
    cout << " Average = " << average <<endl;
    return 0;
}
```

output

sum=154
Average = 30.8


netask2] - Code::Blocks 25.03
Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings
Debug
main() : int
main.cpp main.cpp main.cpp *main.cpp
#include <iostream>
using namespace std;
int main() {
int v1 = 28, v2 = 32, v3 = 37, v4 = 24, v5 = 33;
int sum;
double average;
sum = v1 + v2 + v3 + v4 + v5;
average = sum / 5.0;
cout << "Sum = " << sum << endl;
cout << "Average = " << average << endl;
return 0;
}
& others
Search results
Cccc
Build log
Build messages


ain.cpp [lab3 hometask2] - Code::Blocks 25.03
Edit View Search Project Build Debug Fortran wxSr
"C:\Users\admin\Documents\MyFolder\lab3 hometask
Sum = 154
Average = 30.8
Process returned 0 (0x0) execution tim
Press any key to continue.